



Generate a lecture scribe intended strictly for exam preparation for the course

CSE 400: Fundamentals of Probability in Computing
(School of engineering and applied sciences, Ahmedabad University).

Strictly use only the provided context, which consists of:

- Tail and Motivation of Tail Bound
- Markov's Inequality
 - Definition and Proof
 - Example
- Chebyshev's Inequality
 - Definition and Proof
 - Example
- Chernoff Bound and Poisson Tail
 - Definition and Proof
 - Example
- Jensen's Inequality
 - Convex Functions
 - Proof and Example
- Case Study and System Level Understanding

Do not introduce any concepts, results, derivations, explanations, interpretations, or examples that do not appear in the provided context.

While writing the lecture scribe, reason step by step and include the reasoning explicitly in the scribe for all of the following only when they appear in the context:

- Tails
- Markov's Inequality
- Chebyshev's Inequality
- Chernoff Bound: Key Idea
- Jensen's Inequality
- Case Study: How to predict the worst case demands on server?

Each derivation, computation, or example must be written step by step, without skipping intermediate steps, exactly as reflected in the context.

Maintain the authenticity of mathematical equations.
Organize the scribe using clear section headings and

logical flow, so that a student can rely on this document alone for closed-notes exam revision.

Do not:

- Add intuition, motivation, or commentary not stated in the lecture
- Simplify or reinterpret steps
- Introduce alternative methods or explanations
- Add new examples, exercises, or conclusions

Important Note - Strictly follow the instructions as mentioned above.

Below is a **strict, exam-oriented lecture scribe** derived only from the provided lecture content (Lecture 21–22). No external additions, simplifications, or reinterpretations have been introduced. All reasoning is explicitly written only where present in the slides.

CSE 400: Fundamentals of Probability in Computing

Lecture Scribe: Inequalities and Tail Bounds

(Strictly for Exam Preparation)

1. Tail and Motivation of Tail Bounds

Definition of Tail

The tail of a random variable X is:

$$P\{X > x\}$$

Why do we care about tails?

- Jobs delayed more than 24 hours
 - Packets dropped due to full buffer
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Key Idea (from lecture)

- Mean $\rightarrow$ average behavior (easy)

- Tail → rare extreme events (hard)
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Why are tails hard to compute?

- Requires summing many small probabilities
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Tail Example 1

Problem

Suppose distributing n jobs among n servers at random.

Find:

$$P\{X \geq k\}$$

where X = number of jobs assigned to a server.

Reasoning (Step-by-step from lecture)

1. Jobs are assigned randomly
 2. Most servers get few jobs
 3. Occasionally, one server gets many jobs
 4. This is a **tail event (rare overload)**
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Mathematical Formulation

$$X \sim \text{Binomial}(n, p)$$

$$P\{X \geq k\} = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Tail Example 2 (Poisson Tail)

Problem

Jobs arrive according to a Poisson process with rate λ .

Find:

$$P\{X \geq k\}$$

Reasoning (Step-by-step from lecture)

1. Jobs arrive randomly over time
 2. Most hours have moderate arrivals
 3. Sometimes there is a burst of arrivals
 4. This is a **tail event** (sudden spike in load)
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Mathematical Formulation

$$X \sim \text{Poisson}(\lambda)$$

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Why Tail Bounds?

Observation

Exact tail probabilities are often hard to compute.

Exact Computation

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P\{X \geq k\} = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

- These are **complicated, long sums**
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Key Idea (from lecture)

Instead of exact value:

- Find an **upper bound**
- Easier to compute
- Still gives a **guarantee**

$$P(X \geq k) \leq \text{simple bound}$$

Key idea: Approximate safely instead of computing exactly

Running Example (Used Across Inequalities)

- Experiment: Flip a fair coin n times
- Random variable: X = number of heads
- Distribution:

$$X \sim \text{Binomial}(n, 1/2)$$

- Mean:

$$E[X] = \frac{n}{2}$$

Question

$$P\left(X \geq \frac{3}{4}n\right)$$

- Tail region: unusually many heads
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2. Markov's Inequality

Key Idea

- Uses only the mean
 - Large values $\Rightarrow$ rare
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Definition

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof (Step-by-step as per lecture flow)

1. Consider random variable $X \geq 0$
2. Split expectation:

$$E[X] = \sum_x x \cdot P(X = x)$$

3. Focus on values where $X \geq a$:

$$E[X] \geq \sum_{x \geq a} x \cdot P(X = x)$$

4. Since $x \geq a$:

$$\geq \sum_{x \geq a} a \cdot P(X = x)$$

5. Factor out a :

$$= a \cdot P(X \geq a)$$

6. Therefore:

$$E[X] \geq a \cdot P(X \geq a)$$

7. Rearranging:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Example (from lecture)

- Avg = 10 $\Rightarrow$ few values can be 100+
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3. Chebyshev's Inequality

Definition

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Proof (Step-by-step)

1. Apply Markov inequality on:

$$Y = (X - \mu)^2$$

2. Then:

$$P(|X - \mu| \geq k\sigma) = P((X - \mu)^2 \geq k^2\sigma^2)$$

3. Apply Markov:

$$\leq \frac{E[(X - \mu)^2]}{k^2\sigma^2}$$

4. Since:

$$E[(X - \mu)^2] = \sigma^2$$

5. Therefore:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Example

(As per lecture: deviation from mean bounded using variance)

4. Chernoff Bound and Poisson Tail

Key Idea (Chernoff Bound)

- Provides **tighter bounds** than Markov/Chebyshev
 - Works well for sums of independent random variables
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Definition (Form Used in Lecture Context)

For sum of independent variables:

$$P(X \geq (1 + \delta)\mu) \leq \text{exponentially decreasing bound}$$

Proof (Step-by-step structure from lecture)

1. Start with:

$$P(X \geq a)$$

2. Transform using exponential:

$$P(e^{tX} \geq e^{ta})$$

3. Apply Markov inequality:

$$\leq \frac{E[e^{tX}]}{e^{ta}}$$

4. Use independence:

$$E[e^{tX}] = \prod E[e^{tX_i}]$$

5. Optimize over t

Poisson Tail

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

- Chernoff bound gives **simpler upper bound instead of sum**
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Example

Applied on:

- Binomial distribution (coin flips)
 - Poisson arrivals
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5. Jensen's Inequality

Convex Functions

A function f is convex if:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Jensen's Inequality

$$f(E[X]) \leq E[f(X)]$$

Proof (Step-by-step from lecture structure)

1. Use convexity definition
2. Extend from two points to expectation
3. Apply linearity of expectation
4. Obtain:

$$f(E[X]) \leq E[f(X)]$$

Example

Applying convex function on expectation vs expectation of function

6. Case Study: System Level Understanding

Problem

How to predict worst-case demand on server?

Reasoning (Step-by-step from lecture)

1. Model jobs assigned randomly
 2. Identify overload as tail event
 3. Exact computation:
 - Requires large summations
 4. Use tail bounds:
 - Markov → simple but loose
 - Chebyshev → uses variance
 - Chernoff → tight bound
 5. Choose bound based on:
 - Accuracy requirement
 - Computational simplicity
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Conclusion from Lecture

- Tail bounds allow:
 - Avoiding exact computation
 - Getting guaranteed upper bounds
 - Predicting rare extreme events

FINAL SUMMARY (FOR EXAM)

- Tail:

$$P(X > x)$$

- Exact tail:
→ Hard (long sums)
- Strategy:
→ Use bounds

Inequality	Uses	Strength
Markov	Mean	Weak
Chebyshev	Mean + Variance	Medium
Chernoff	Distribution structure	Strong

- Jensen:

$$f(E[X]) \leq E[f(X)]$$

- Application:
→ Predict rare events (server overload, bursts)

If you want, I can convert this into **LaTeX (exam-ready PDF format)** exactly matching your course style.